

Remark 5.5. While it suffices to lower bound the strictest term $\text{SDA}(\mathcal{P}; \text{nb}(\mathcal{M}))$, we exhibit different constructions for each of the three dimensions as they offer increasing levels of generality. In particular, $\text{SDA}(\mathcal{P}; \emptyset)$ can be realized by $\mathcal{P}' \subset \mathcal{P}$ containing any prescribed $p \in \mathcal{P}$ and for any $\mathcal{D}$. Moreover, the difficulty is solely due to the logical part of the task; without pretraining, the reasoner will take exponentially many guesses to even produce a valid path.

Corollary 5.6 (hardness without global search). *Suppose $f_\theta(\text{nb}(\mathcal{M}_p(X_{\text{in}}, X_{\text{out}})), \mathcal{A})$ is any parametric model with polynomially bounded gradients, that can freely search a local neighborhood of the generated CoT. Then any iterative algorithm $A(\theta)$ that makes at most polynomial queries to the $e^{-\Omega(K)}$ -corrupted gradient oracle ∇L satisfies*

$$\mathbb{E}_p[L(A(\theta); p)] \geq 1 - e^{-\Omega(K)},$$

with probability $1 - e^{-\Omega(K)}$ for M sufficiently large.

Hence $\mathcal{H}$ cannot be even weakly learned in polynomial time if search is not long enough. The key intuition is that if the graph is locally isomorphic, local search cannot distinguish between sparse inter-cluster edges and low-probability but within-cluster edges as it cannot explore the whole cluster. This demonstrates the importance of spending sufficient inference-time compute for improving reasoning ability.

6 Conclusion

We introduced a metastable Markov framework for modeling CoT reasoning, revealing the benefits of inference-time search, RL, and distillation. We showed that search can improve reasoning by identifying critical sparse transitions (hard steps), which can then be leveraged to fine-tune the pretrained model via RL or distilled into a more efficient representation, improving hitting times for path generation. We further established learning-theoretic limits on reasoning with restricted information and showed that logical reasoning tasks become intractable without global search.

Future directions. We have studied a simple curiosity-based unsupervised reward model; it would be interesting to see how a more complex search process could be guided with outcome rewards. Our framework could also be used to study other inference-time methods such as CoT revision (e.g., backtracking to better locate sparse edges), as well as iterative finetuning of the pretrained model, and explore scaling laws for inference time compute.

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References

- Emmanuel Abbe, Samy Bengio, Aryo Lotfi, Colin Sandon, and Omid Saremi. How far can transformers reason? The locality barrier and inductive scratchpad. In *Advances in Neural Information Processing Systems*, 2024.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.
- J. Beltrán and C. Landim. Metastability of reversible finite state Markov processes. *Stochastic Processes and their Applications*, 121(8):1633–1677, 2011.
- Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, et al. Graph of thoughts: solving elaborate problems with large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.
- Volker Betz and Stéphane Le Roux. Multi-scale metastable dynamics and the asymptotic stationary distribution of perturbed Markov chains. *Stochastic Processes and their Applications*, 126(11):3499–3526, 2016.
- Satwik Bhattamishra, Arkil Patel, Phil Blunsom, and Varun Kanade. Understanding in-context learning in transformers and LLMs by learning to learn discrete functions. In *International Conference on Learning Representations*, 2024.
- Alessandra Bianchi and Alexandre Gaudillière. Metastable states, quasi-stationary distributions and soft measures. *Stochastic Processes and their Applications*, 126, 2016.
- Anton Bovier, Michael Eckhoff, Véronique Gayrard, and Markus Klein. Metastability and low lying spectra in reversible Markov chains. *Communications in Mathematical Physics*, 228:219–255, 2002.
- Nader Bshouty and Vitaly Feldman. On using extended statistical queries to avoid membership queries. *Journal of Machine Learning Research*, 2:529–545, 09 2001.
- Yuri Burda, Harri Edwards, Deepak Pathak, Amos Storkey, Trevor Darrell, and Alexei A. Efros. Large-scale study of curiosity-driven learning. In *International Conference on Learning Representations*, 2018.
- Yuri Burda, Harrison Edwards, Amos Storkey, and Oleg Klimov. Exploration by random network distillation. In *International Conference on Learning Representations*, 2019.

- David Chiang, Peter Cholak, and Anand Pillay. Tighter bounds on the expressivity of transformer encoders. In *International Conference on Machine Learning*, 2023.
- Emilio Cirillo, Francesca Nardi, and Julien Sohier. Metastability for general dynamics with rare transitions: escape time and critical configurations. *Journal of Statistical Physics*, 161, 2014.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Ezra Edelman, Nikolaos Tsilivis, Benjamin L. Edelman, Eran Malach, and Surbhi Goel. The evolution of statistical induction heads: in-context learning Markov chains. *Advances in Neural Information Processing Systems*, 2024.
- Konstantin Fackeldey, Alexander Sikorski, and M. Weber. Spectral clustering for non-reversible Markov chains. *Computational and Applied Mathematics*, 37, 2018.
- Vitaly Feldman. A general characterization of the statistical query complexity. *Proceedings of Machine Learning Research*, 65:785–830, 2017.
- Guhao Feng, Bohang Zhang, Yuntian Gu, Haotian Ye, Di He, and Liwei Wang. Towards revealing the mystery behind chain of thought: a theoretical perspective. In *Advances in Neural Information Processing Systems*, 2023a.
- Xidong Feng, Ziyu Wan, Muning Wen, Stephen Marcus McAleer, Ying Wen, Weinan Zhang, and Jun Wang. Alphazero-like tree-search can guide large language model decoding and training. *arXiv preprint arXiv:2309.17179*, 2023b.
- R. Fernandez, F. Manzo, F. R. Nardi, E. Scoppola, and J. Sohier. Conditioned, quasi-stationary, restricted measures and escape from metastable states. *The Annals of Applied Probability*, 26(2): 760–793, 2016.
- Roberto Fernandez, Francesco Manzo, Francesca Nardi, and Elisabetta Scoppola. Asymptotically exponential hitting times and metastability: A pathwise approach without reversibility. *Electronic Journal of Probability*, 20, 2014.
- David Fritzsche, Volker Mehrmann, Daniel Szyld, and Elena Virnik. An SVD approach to identifying metastable states of Markov chains. *Electronic Transactions on Numerical Analysis*, 29: 46–69, 2008.
- Kanishk Gandhi, Denise Lee, Gabriel Grand, Muxin Liu, Winson Cheng, Archit Sharma, and Noah D Goodman. Stream of search (SoS): learning to search in language. *arXiv preprint arXiv:2404.03683*, 2024.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. DeepSeek-R1: incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.